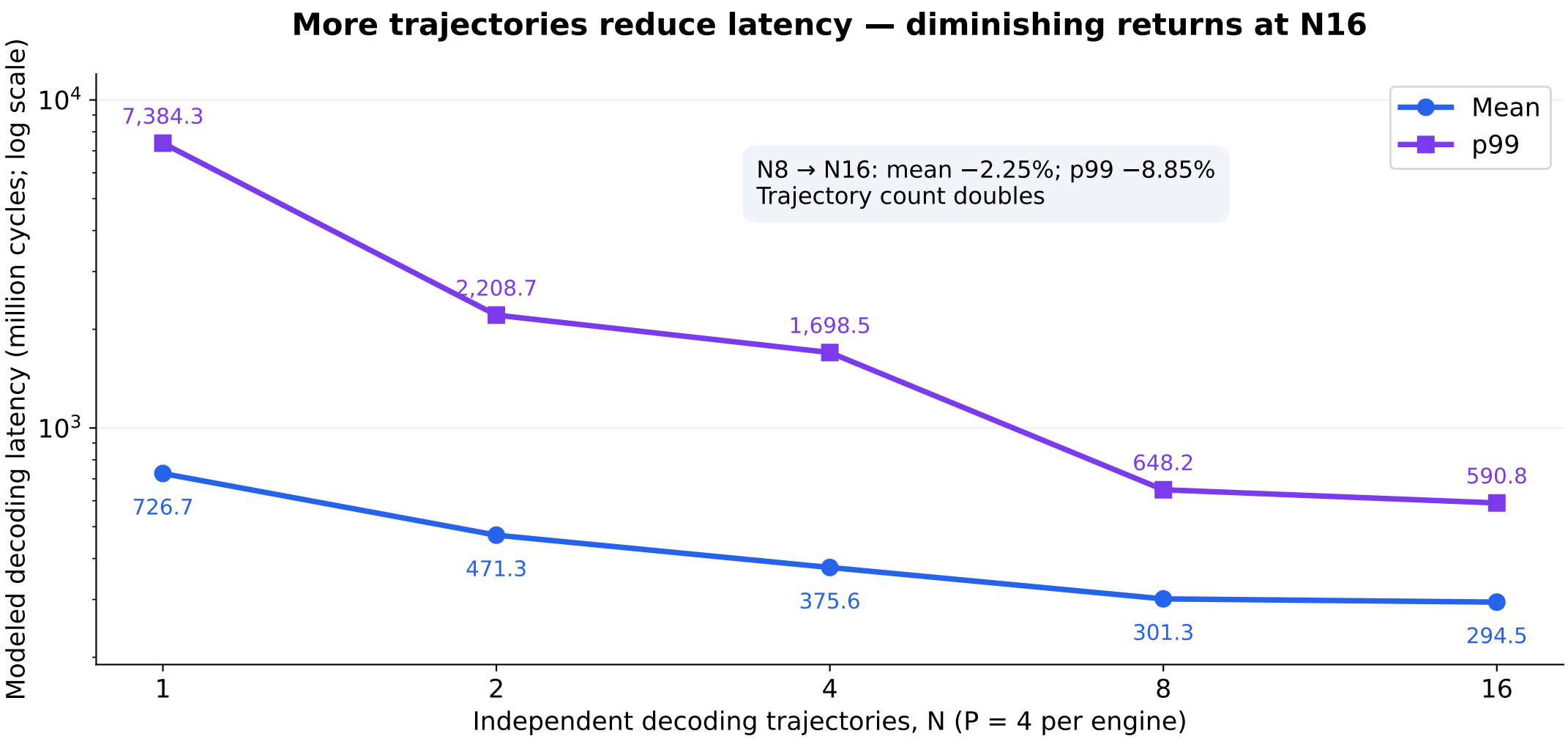

Canonical Gross [[144,12,12]]: 128 paired shots, p = 0.003; mean and linear-interpolated p99.
Modeled first-success / last-engine-on-all-fail latency; not measured FPGA/RTL timing. p99 is tail-sensitive at 128 shots.